

Customer Churn Analysis Project

Comprehensive Exploratory Data Analysis & Documentation

1. Project Objective

The primary mission of this analytical project is to diagnose the root causes of customer churn within the telecom subscriber base. By systematically dissecting demographic data, service usage patterns, and billing structures, we aim to derive actionable intelligence that can arrest attrition rates.

Key Goals:

- Isolate high-risk customer segments.
- Evaluate the impact of service bundling on retention.
- Analyze the efficacy of long-term contracts versus month-to-month plans.
- Assess the correlation between pricing strategies and customer loyalty.

2. Tools Used

The following technical stack was employed to execute this analysis:

- Python 3.x:** Core analytical engine.
- Pandas:** For robust data manipulation and cleaning.
- Matplotlib & Seaborn:** For generating high-fidelity visualizations.
- NumPy:** For high-performance numerical computations.

3. About the Dataset

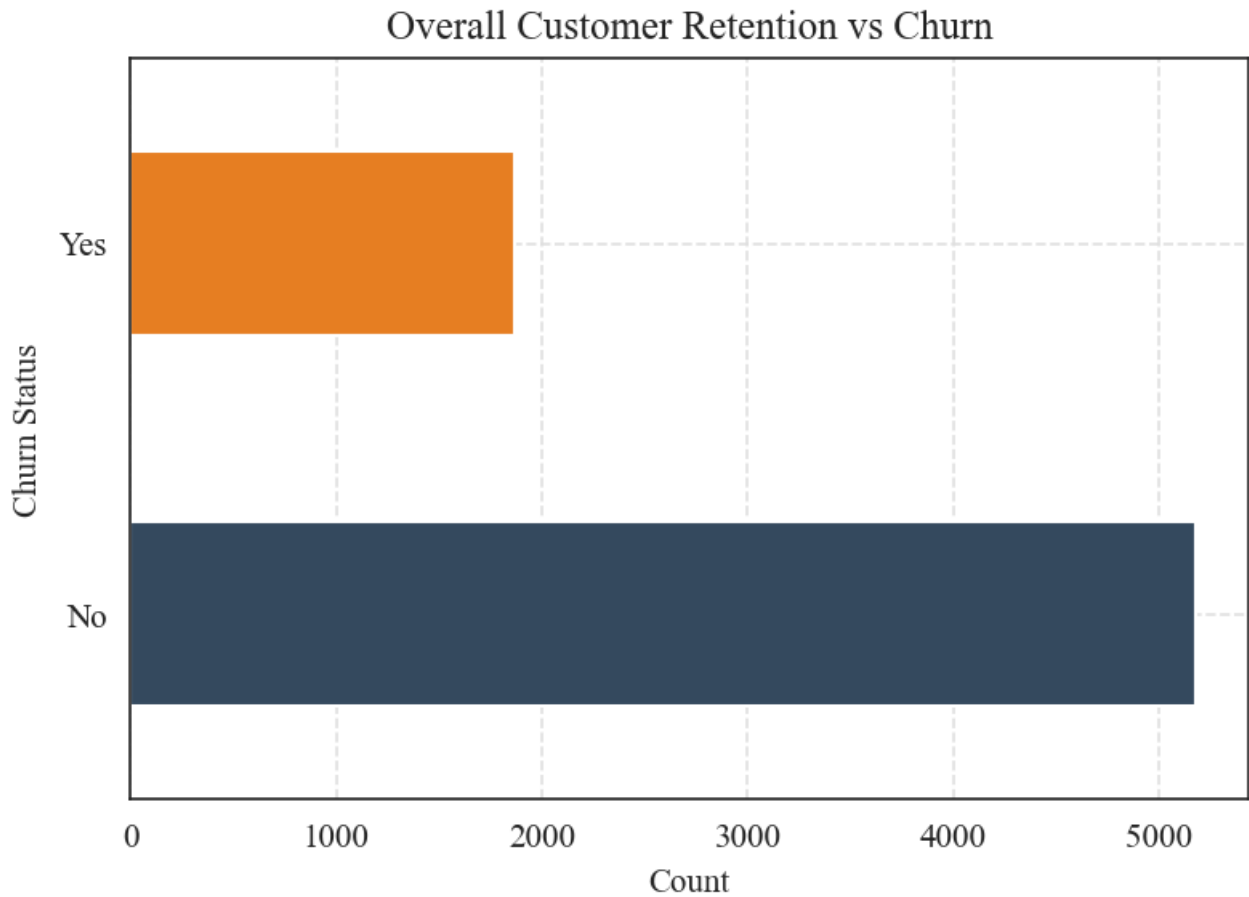
The dataset serves as a comprehensive collection of customer attributes, including:

- Demographics:** Gender, Age (Senior Citizen status), and Geography.
- Account Information:** Tenure, Contract Type, Payment Method, Paperless Billing.
- Service Portfolio:** Phone, Internet (DSL/Fiber), Tech Support, Streaming.
- Charges:** Monthly and Total charges incurred.

4. Visual Analysis & Key Insights

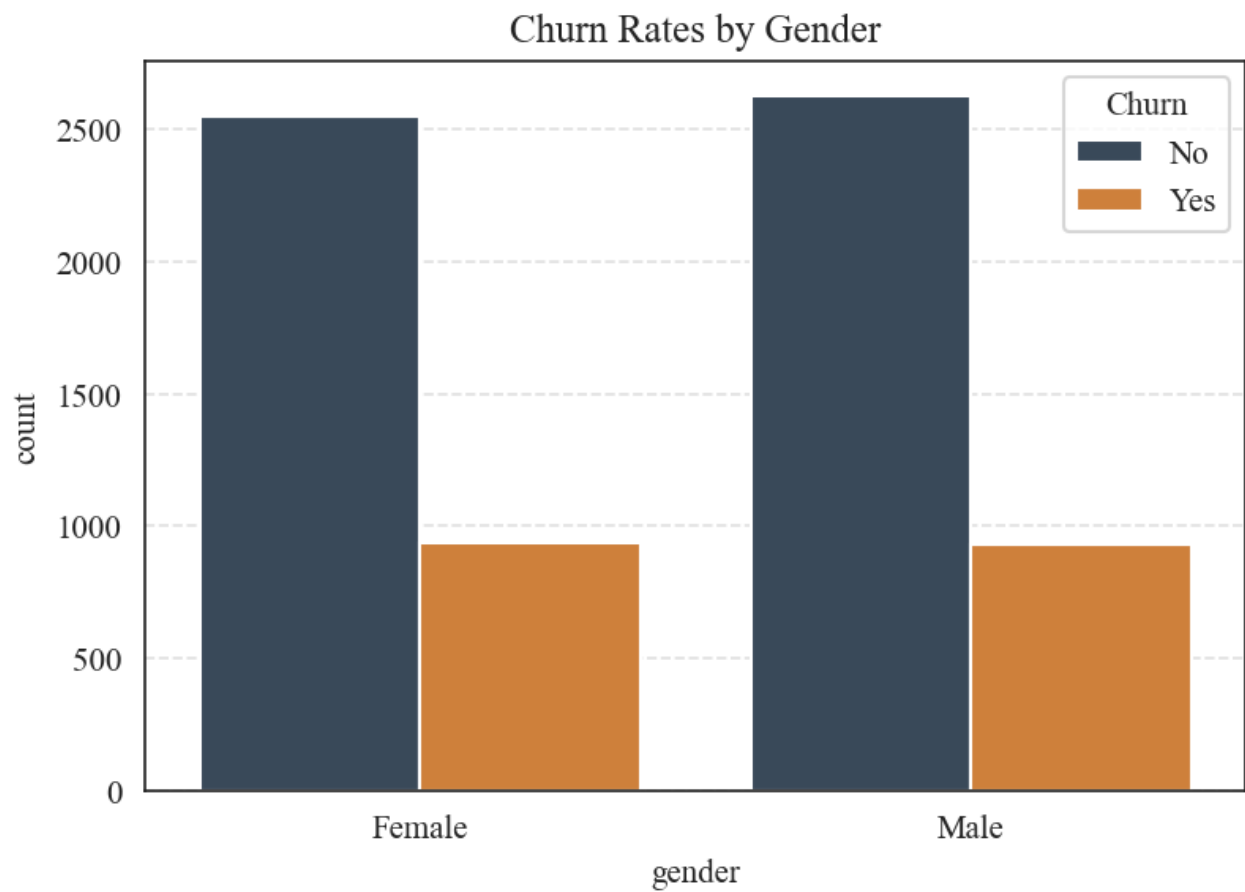
This section details a step-by-step visual investigation into the dataset factors.

1. Overall Churn Count



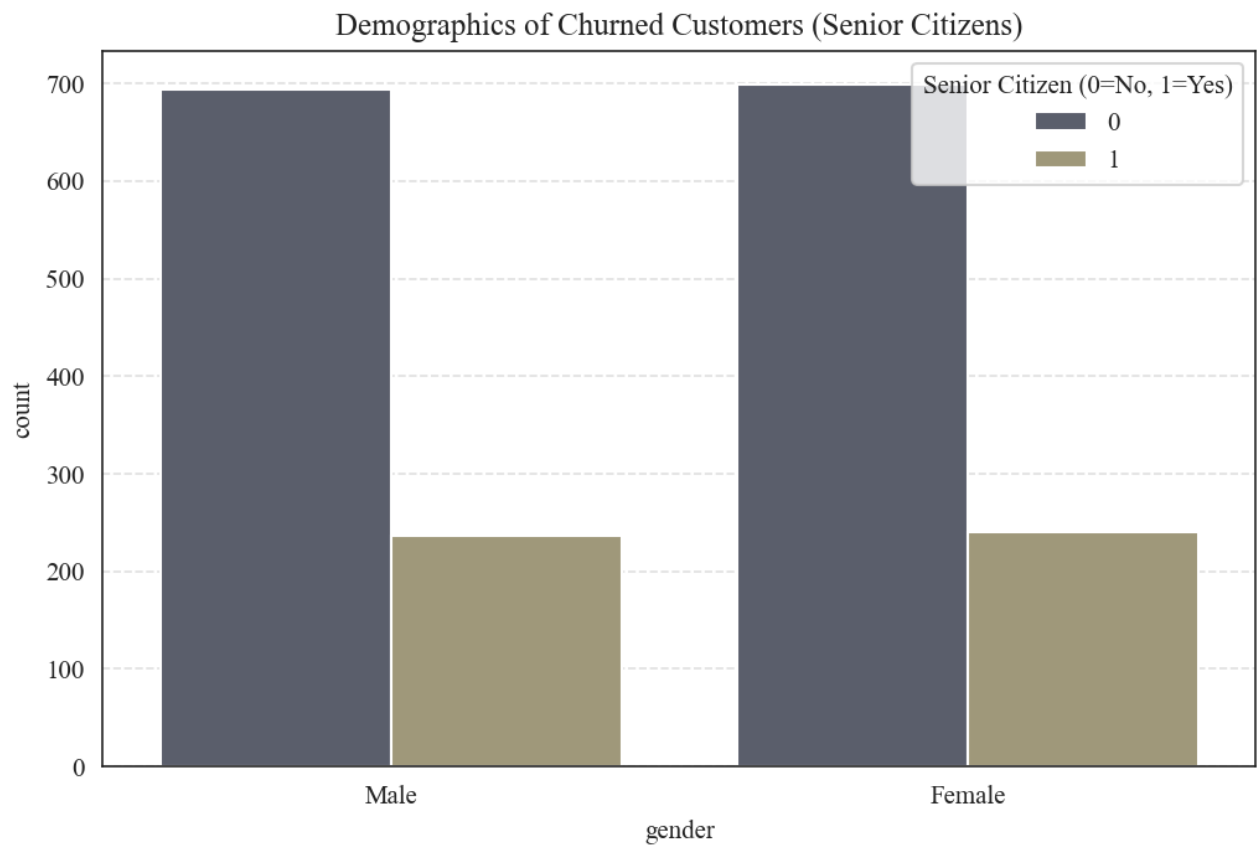
Analysis: The dataset reveals a class imbalance where retained customers significantly outnumber churned ones. However, the churn proportion is substantial enough to warrant a dedicated retention strategy. The horizontal bar chart clearly delineates the volume difference.

2. Churn by Gender



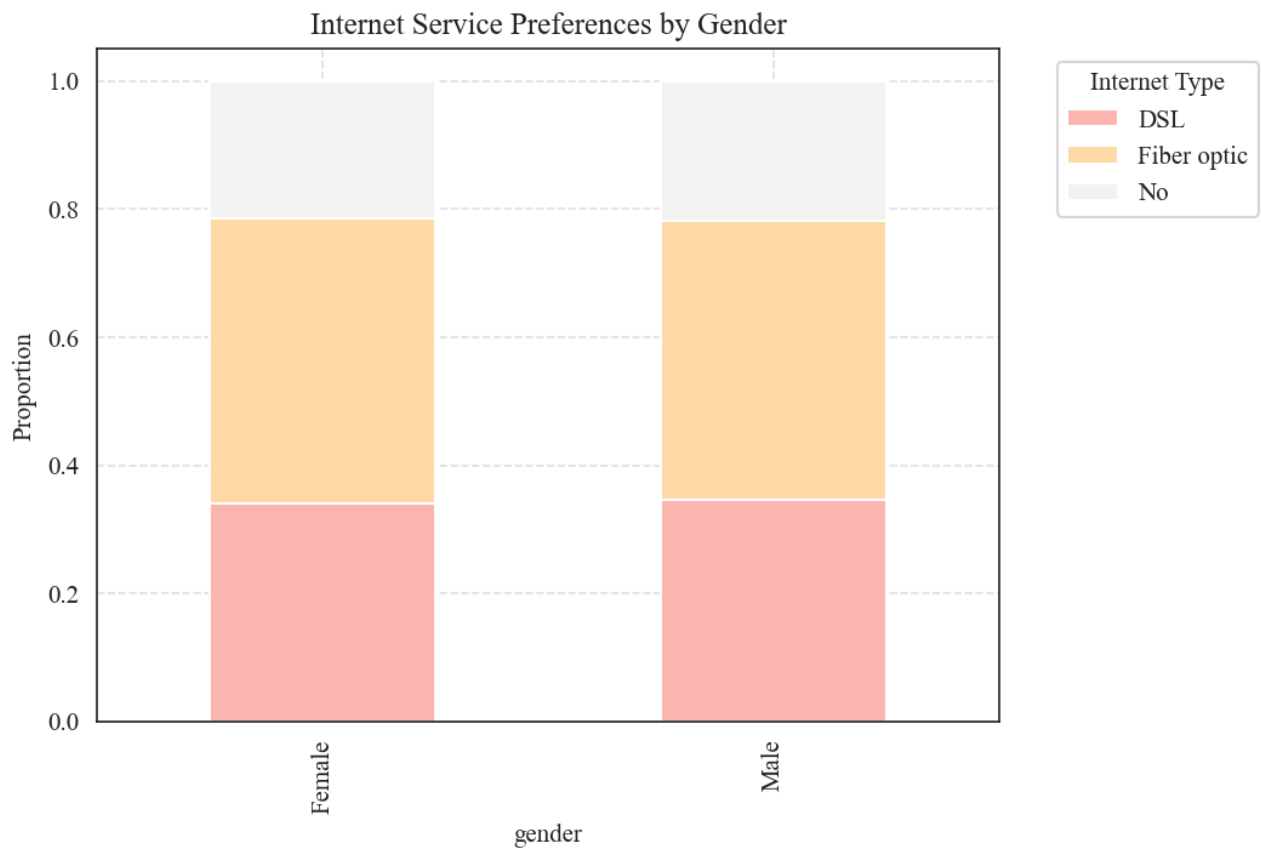
Analysis: Gender appears to be a neutral factor. The churn rates for males and females are statistically indistinguishable, suggesting that demographic targeting based on gender alone would be ineffective.

3. Churn: Gender & Senior Citizens



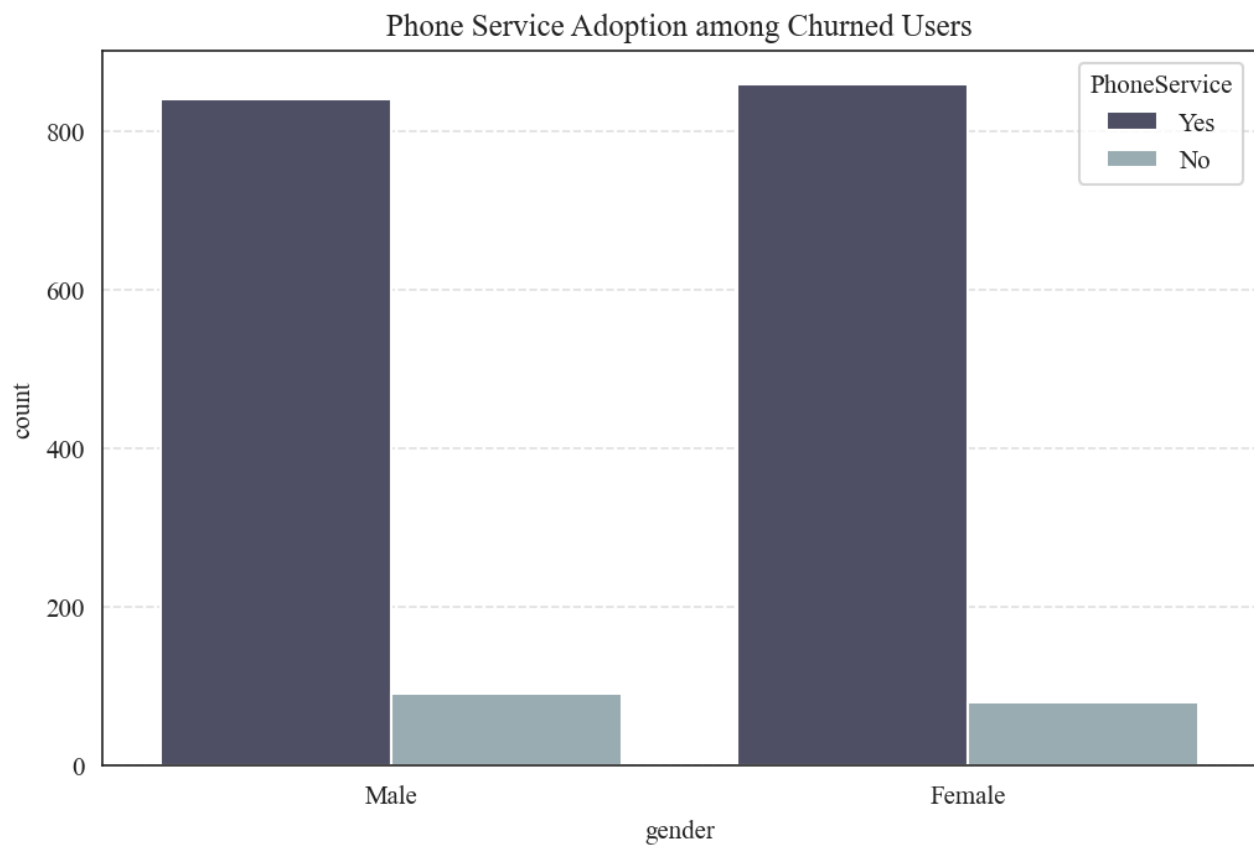
Analysis: Among the churned population, Senior Citizens represent a specific cohort of concern. Their churn behavior differs from the general population, possibly indicating issues with accessibility or pricing structures suited for fixed incomes.

4. Internet Choices by Gender



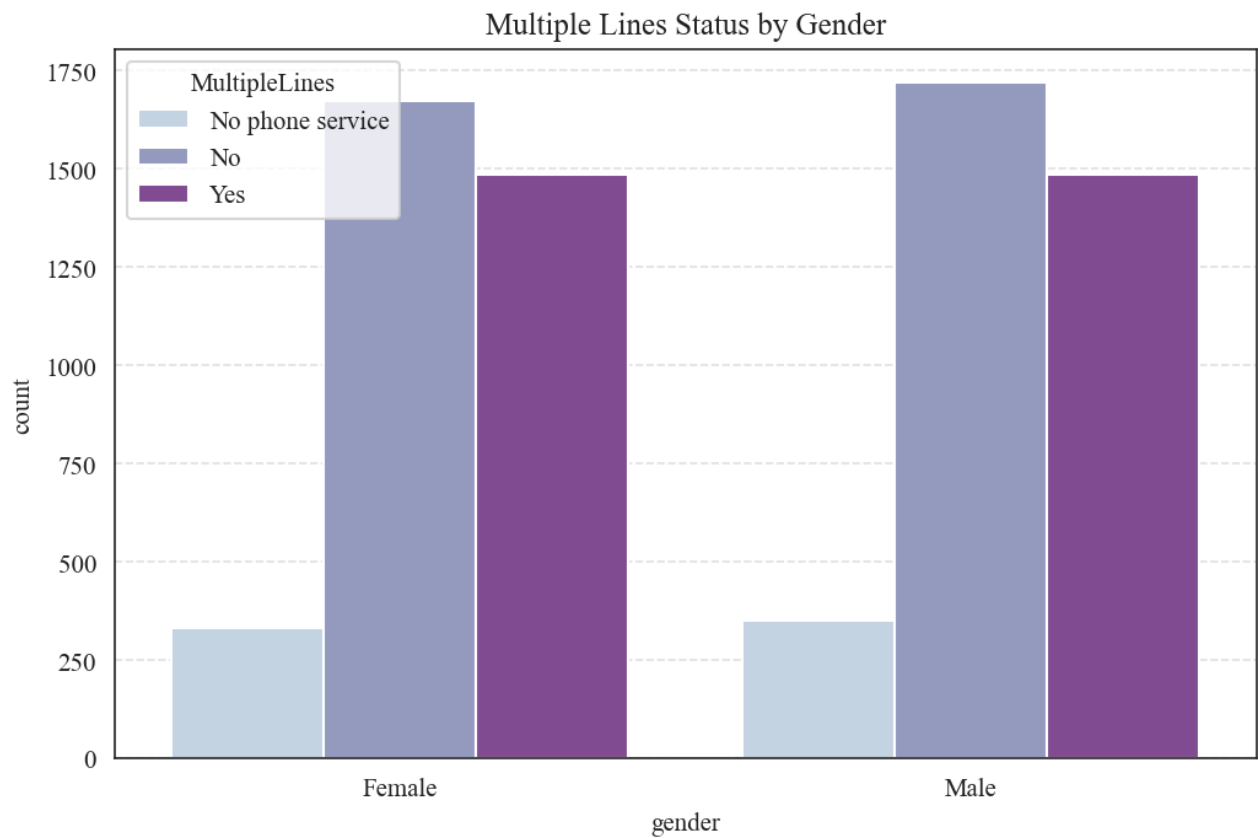
Analysis: The stacked bar chart illustrates the market penetration of various internet types. Fiber Optic has high adoption across genders, but as we will see, it also carries higher churn risks.

5. Phone Service vs Churn



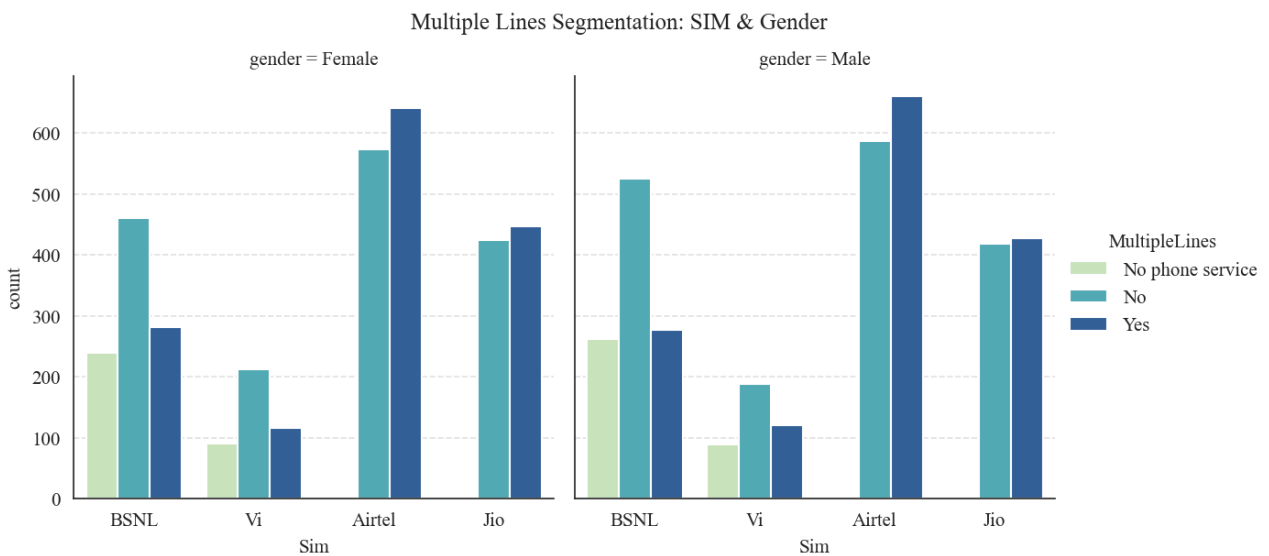
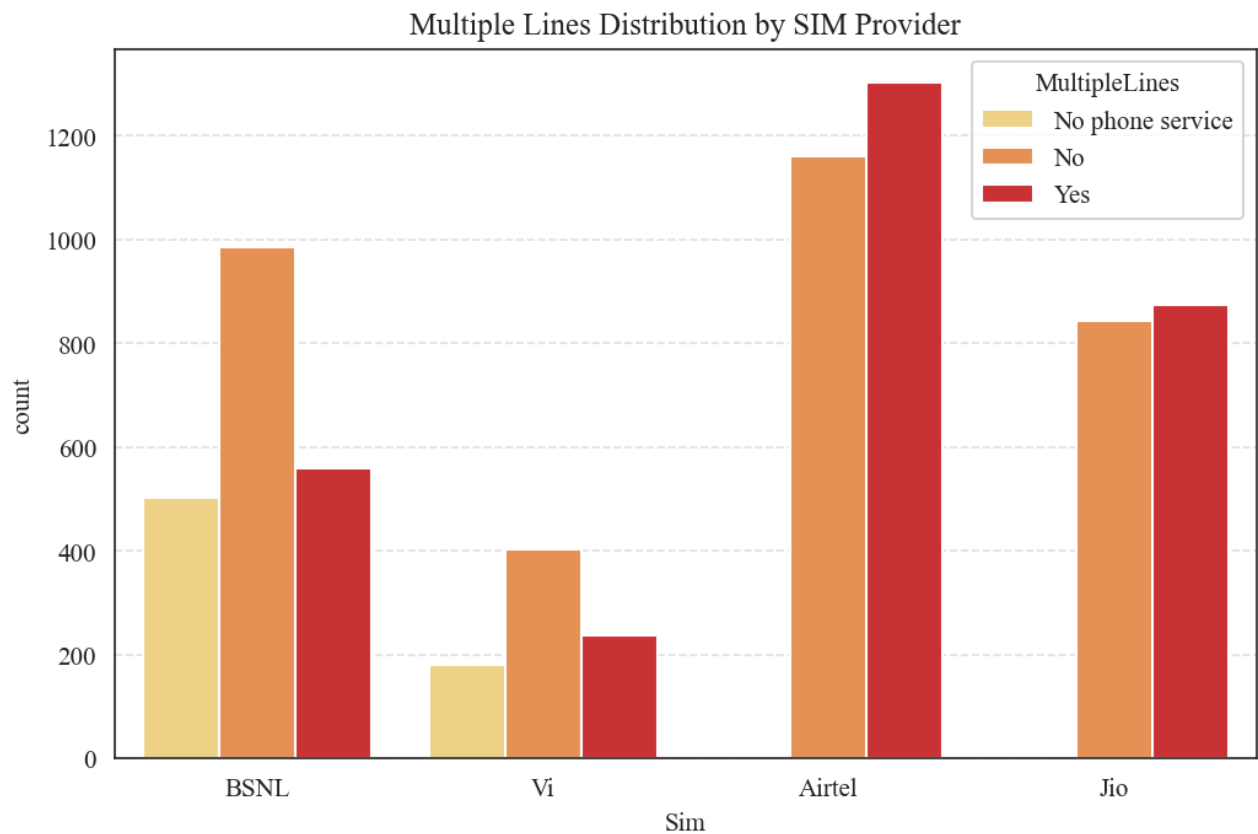
Analysis: The vast majority of churned customers did subscribe to Phone Service. This indicates that having a phone line does not strictly act as a retention anchor.

6. Multiple Lines by Gender



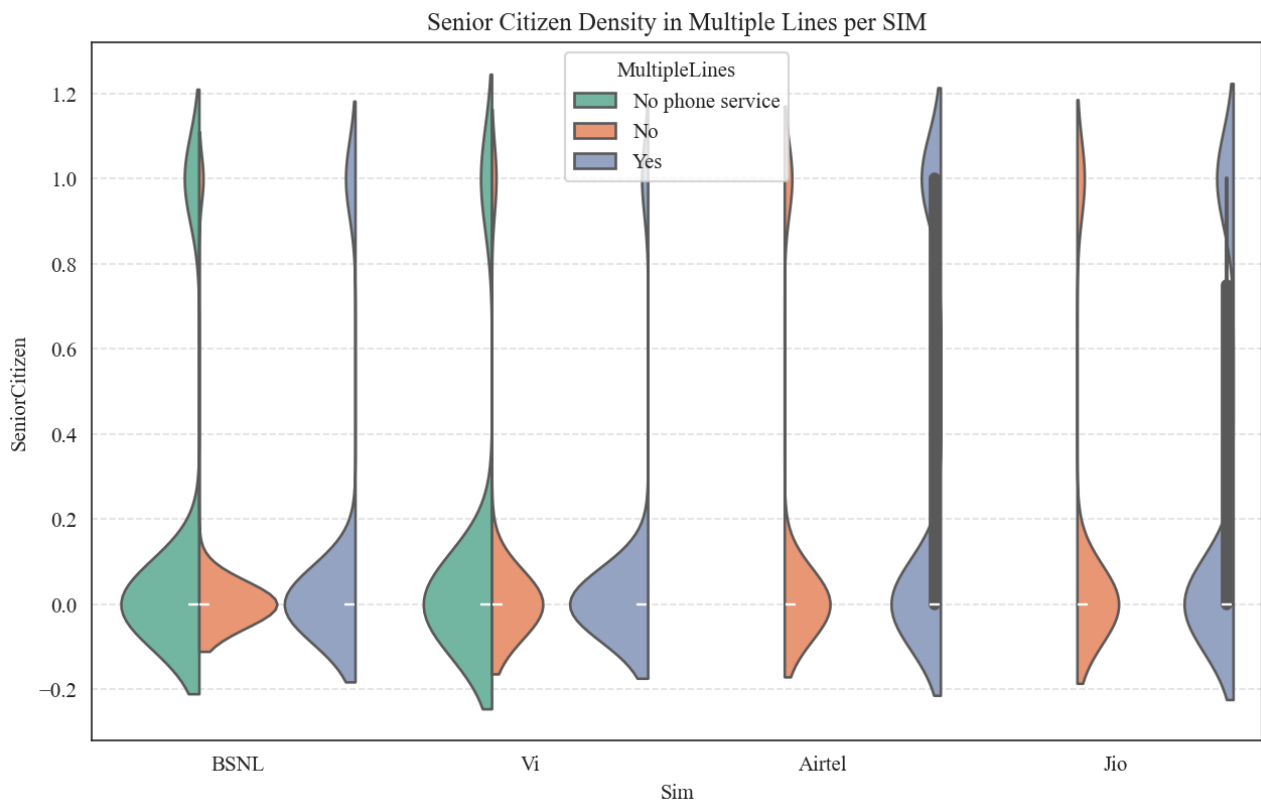
Analysis: Multi-line adoption is symmetric across genders. It does not appear to be a gender-specific preference.

7 & 8. Multiple Lines and SIM Operators



Analysis: We observe distinct variances in multi-line subscriptions when segmented by SIM operator. Certain operators seem to bundle multiple lines more effectively.

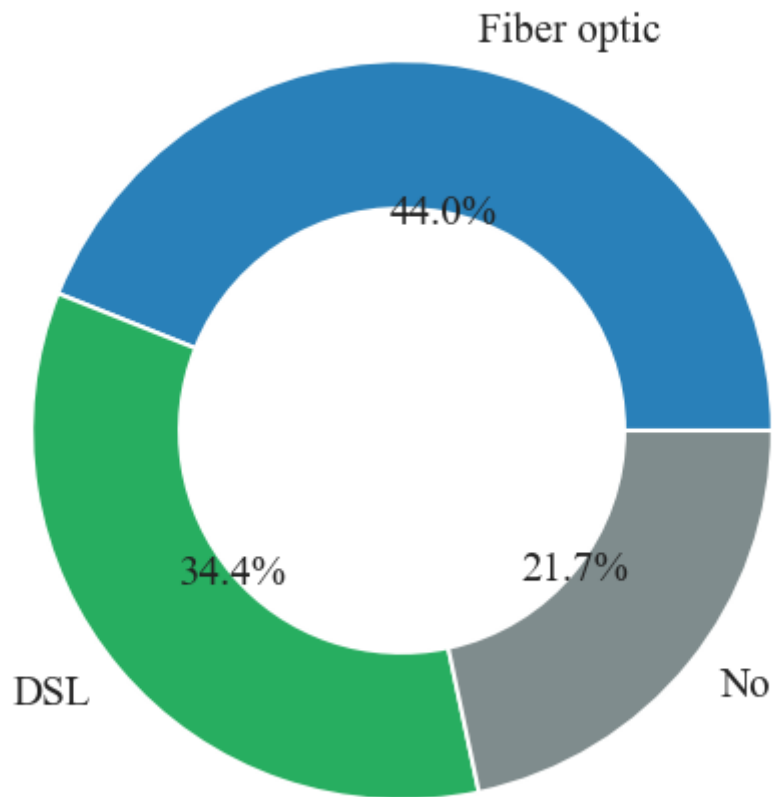
9. Multiple Lines & Senior Citizens



Analysis: The violin plot highlights that Senior Citizens are less likely to manage or pay for multiple lines, preferring simpler, single-line connectivity.

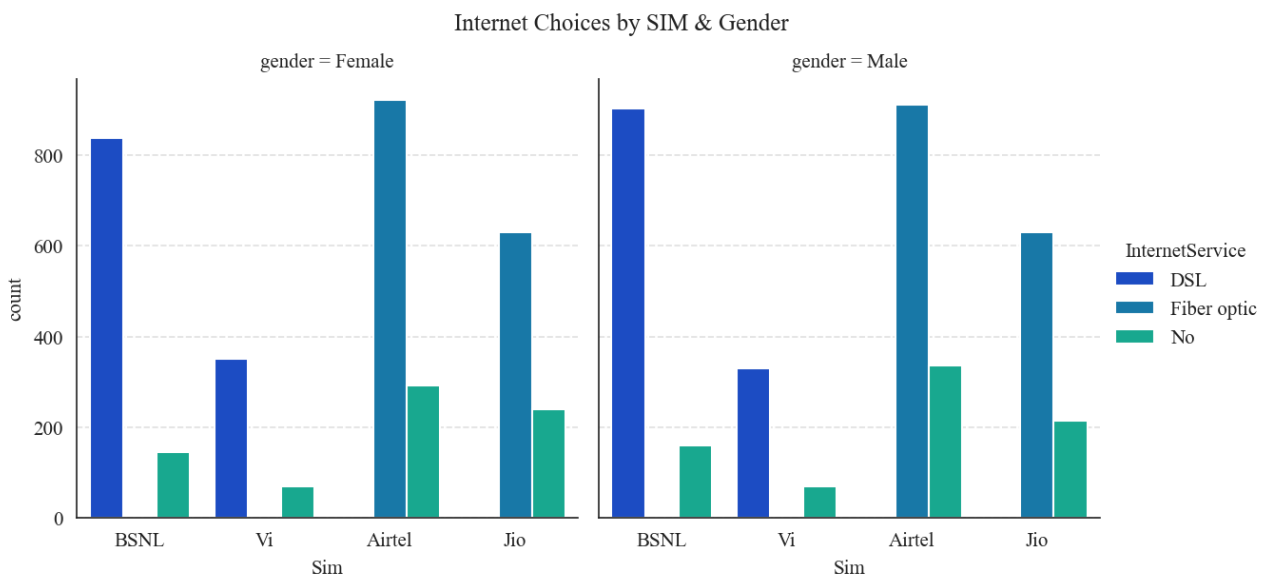
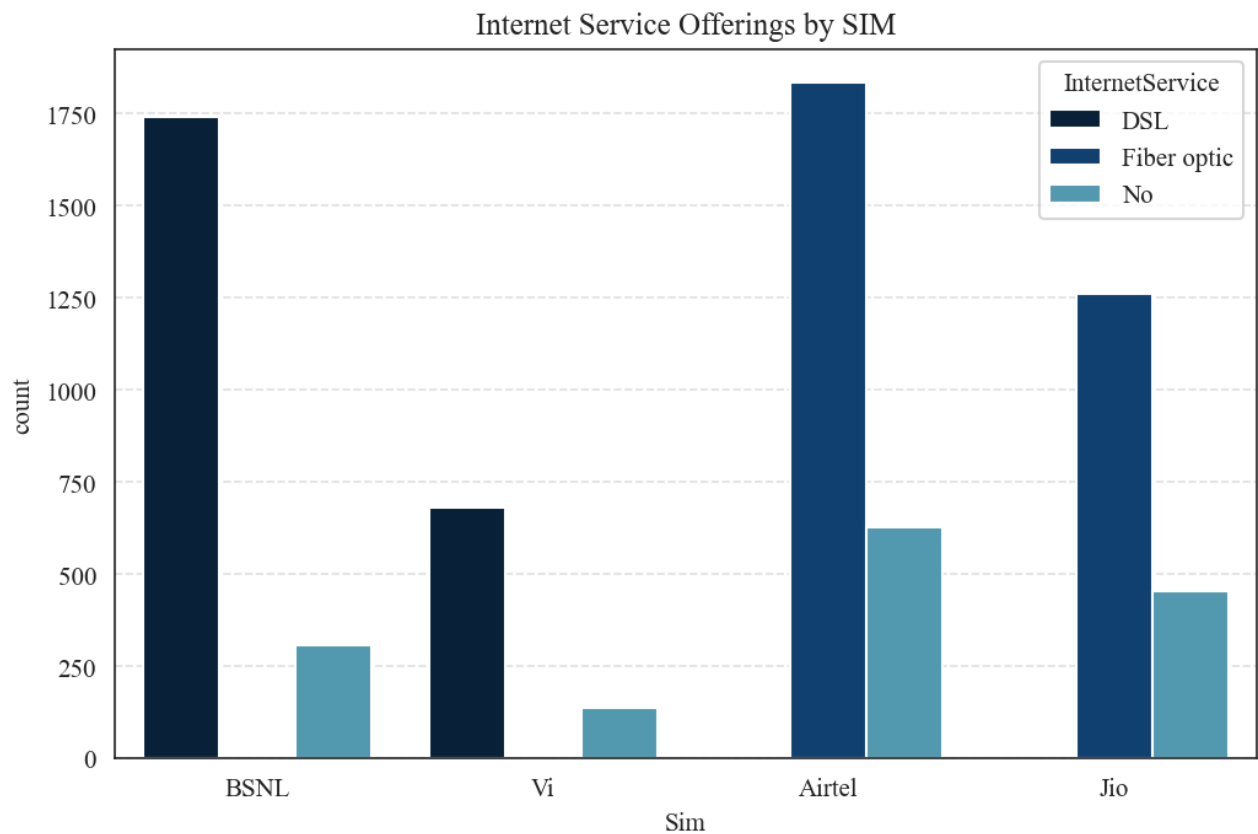
10. Internet Service Popularity

Market Share of Internet Services



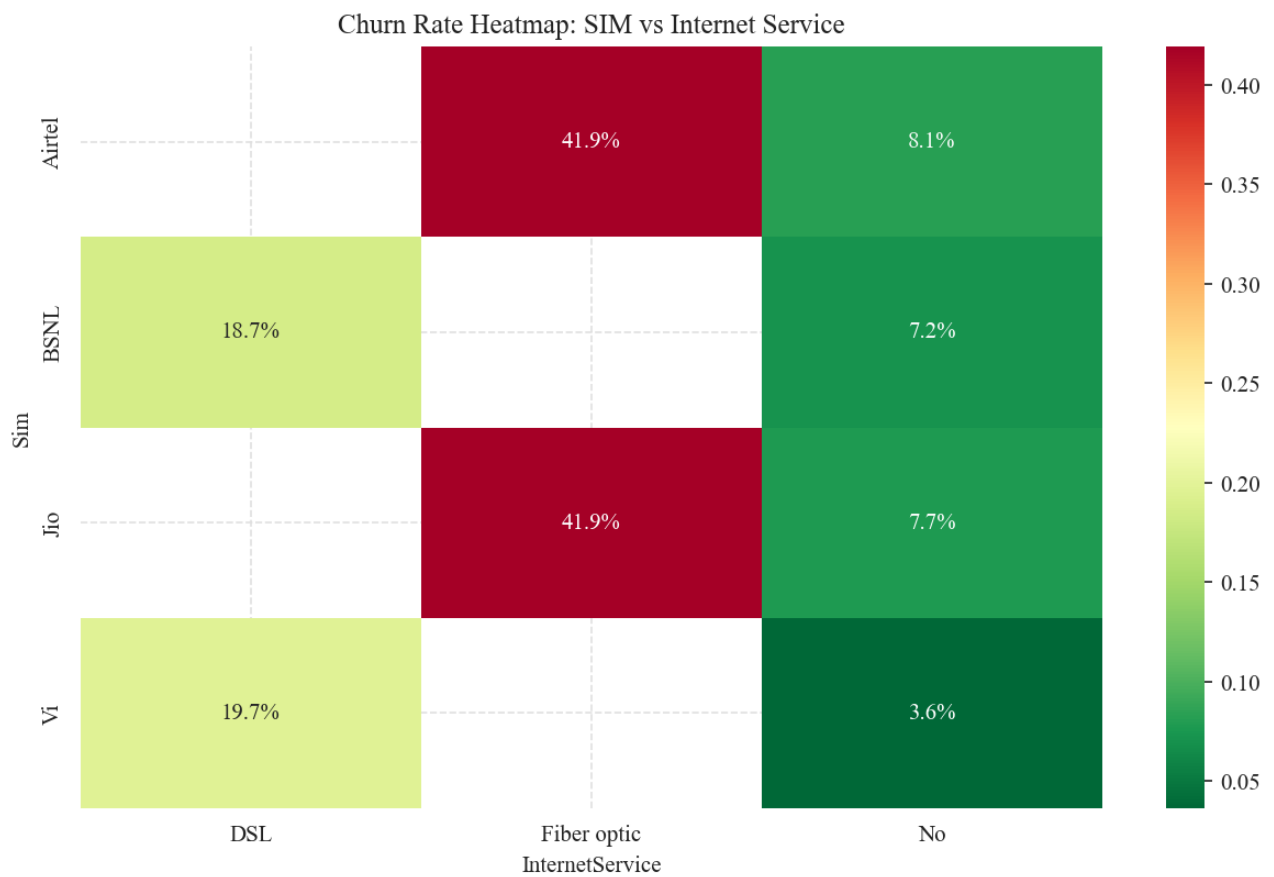
Analysis: The donut chart visualizes market share. Fiber Optic dominates, followed by DSL. A small segment retains service without internet, likely for voice-only needs.

11 & 12. Internet Service by Operator



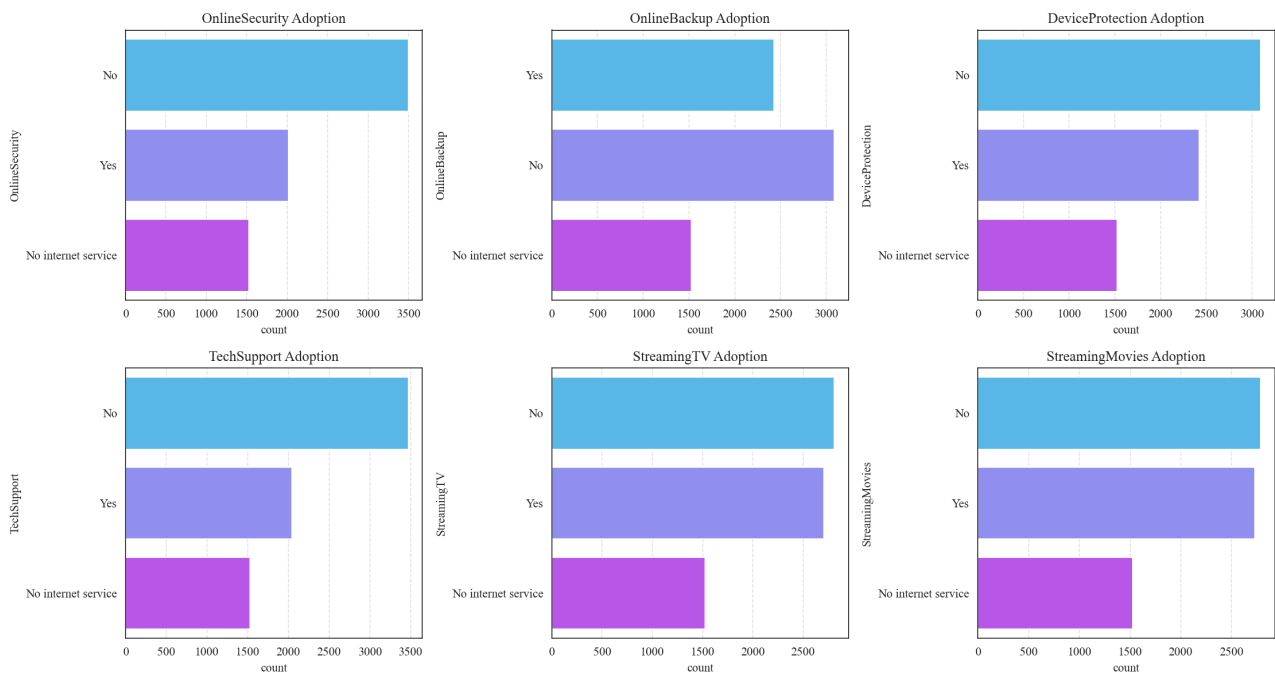
Analysis: Infrastructure availability varies by operator, dictating the internet types available to their respective subscriber bases.

13. Churn by Internet Service



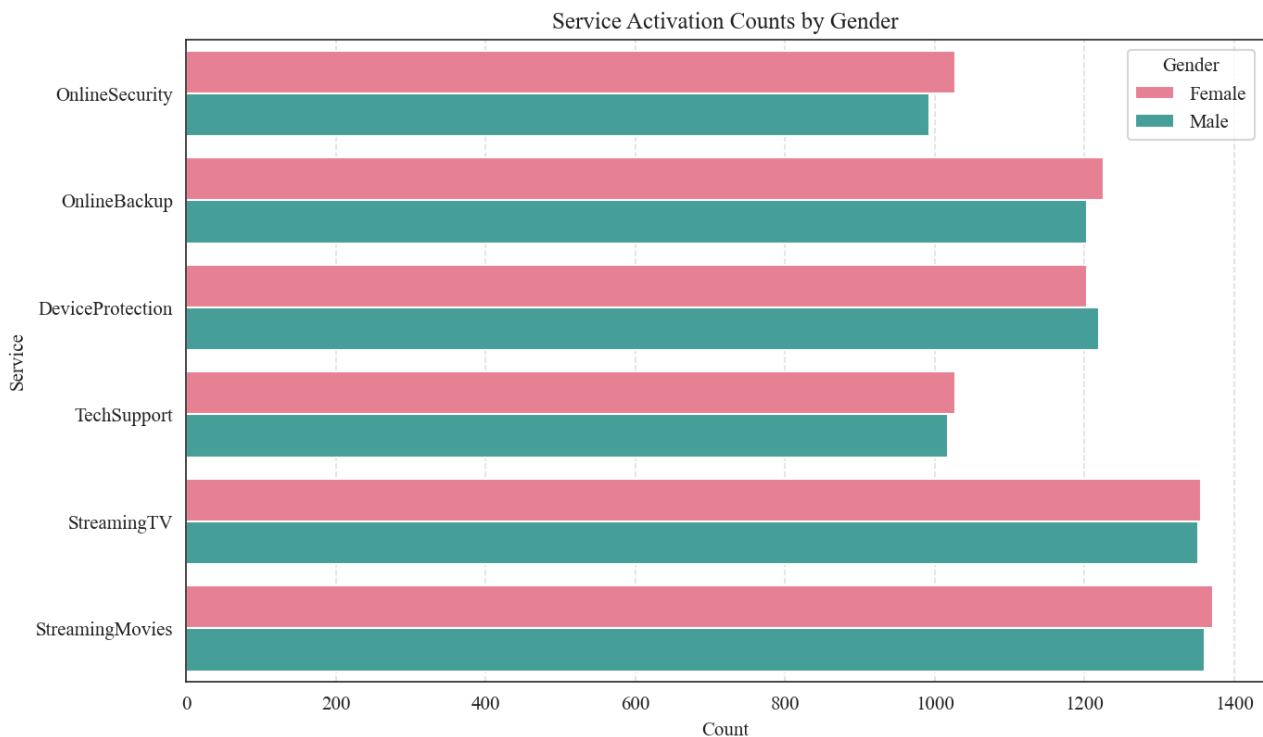
Analysis: The heatmap provides a striking insight: Red zones indicate high churn. Fiber Optic users show alarmingly high churn rates compared to DSL, pointing to potential service quality/reliability issues or price dissatisfaction.

14. Add-on Services



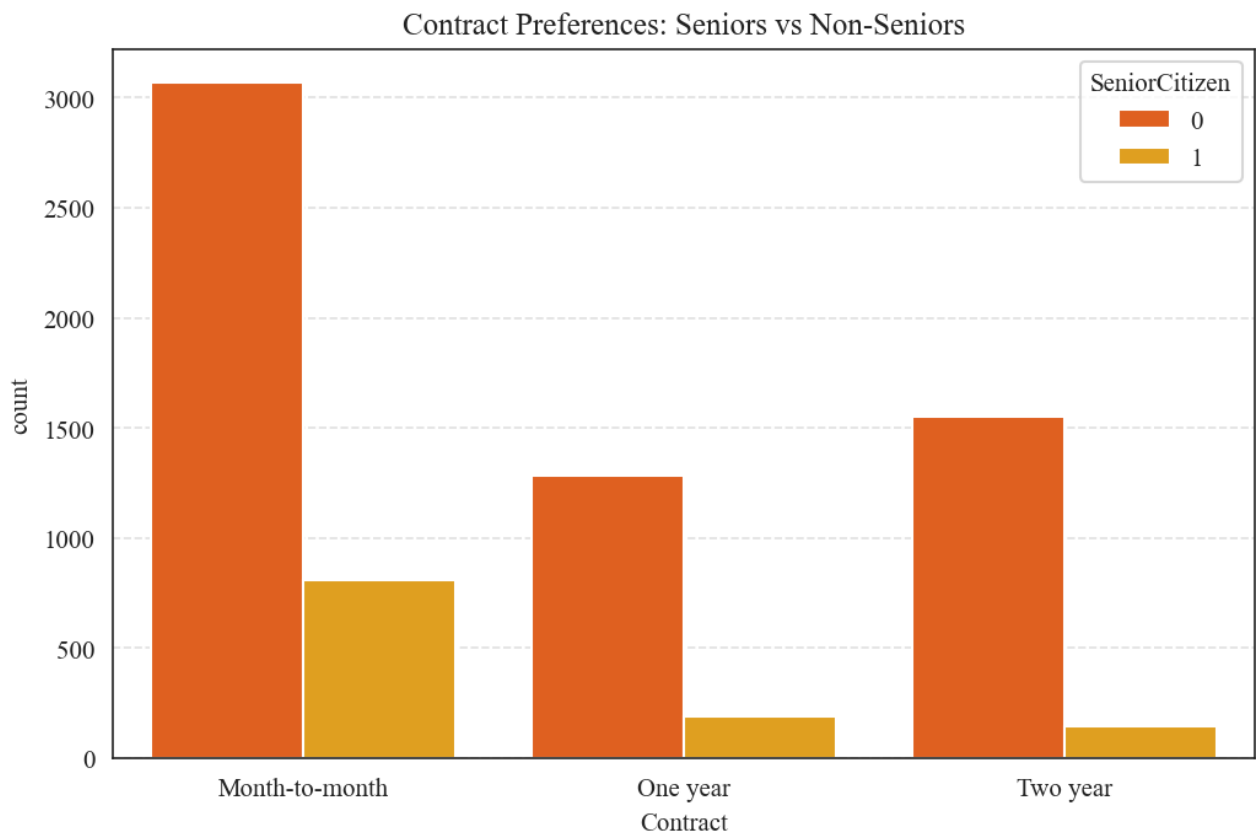
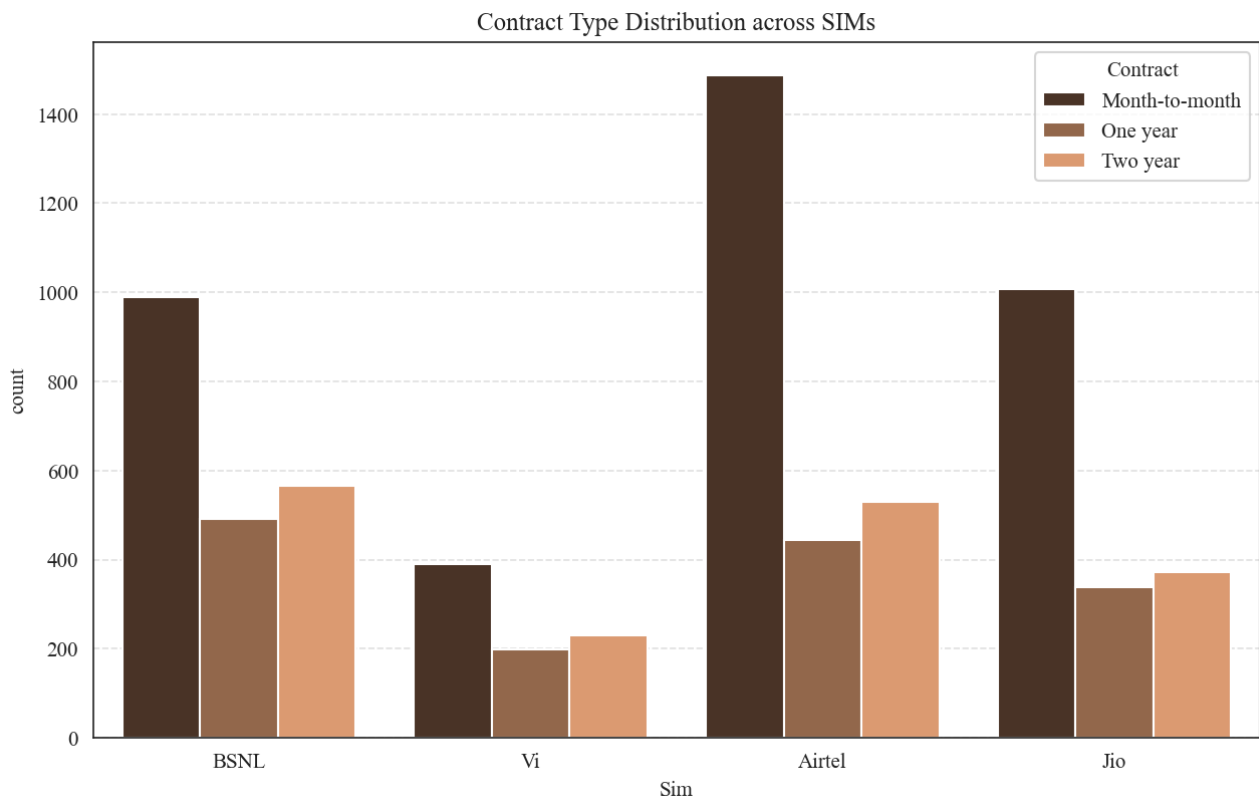
Analysis: Adoption of value-added services like Tech Support and Online Security is generally low. However, these services are crucial for "stickiness".

15. Services by Gender



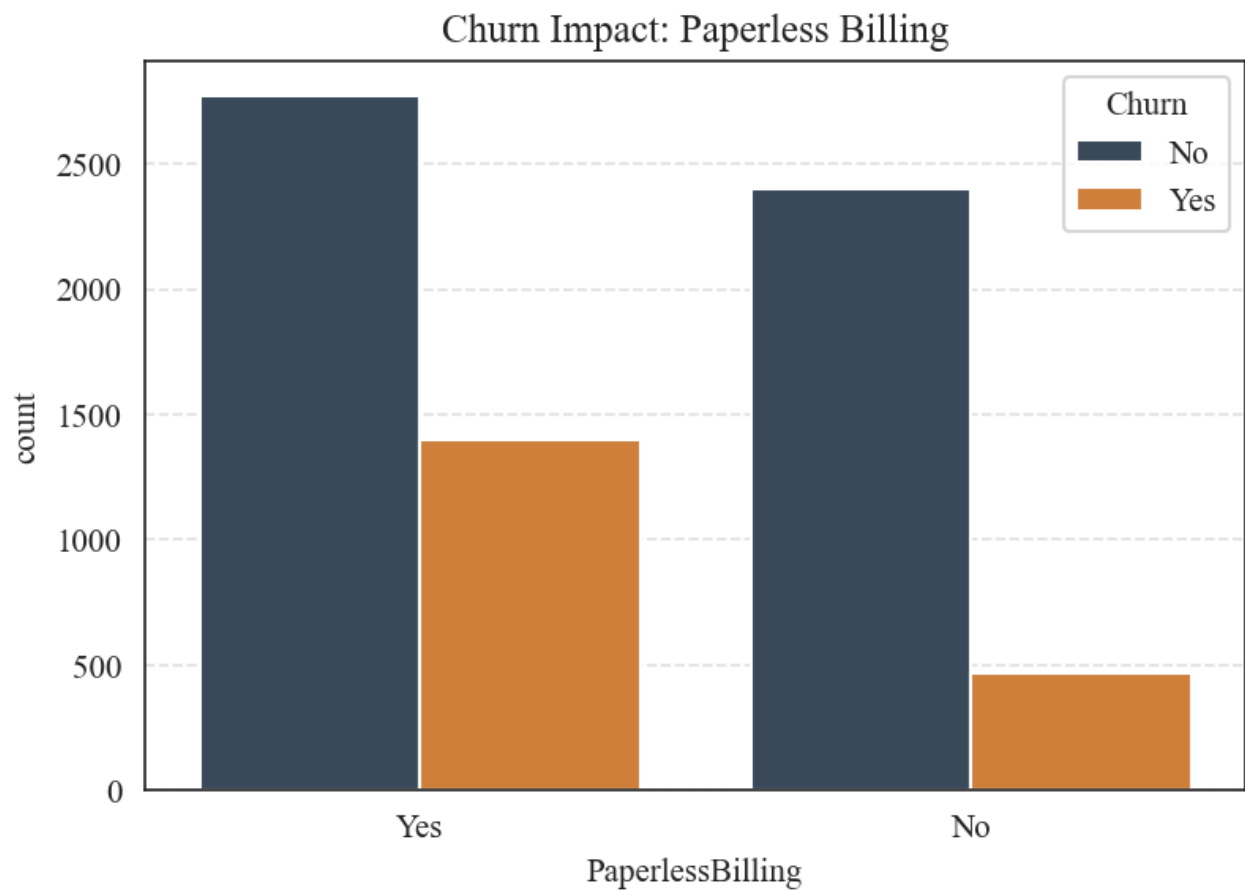
Analysis: Service consumption patterns are remarkably similar between men and women, reinforcing the conclusion that gender is not a primary differentiator in this dataset.

16 & 17. Contract Types



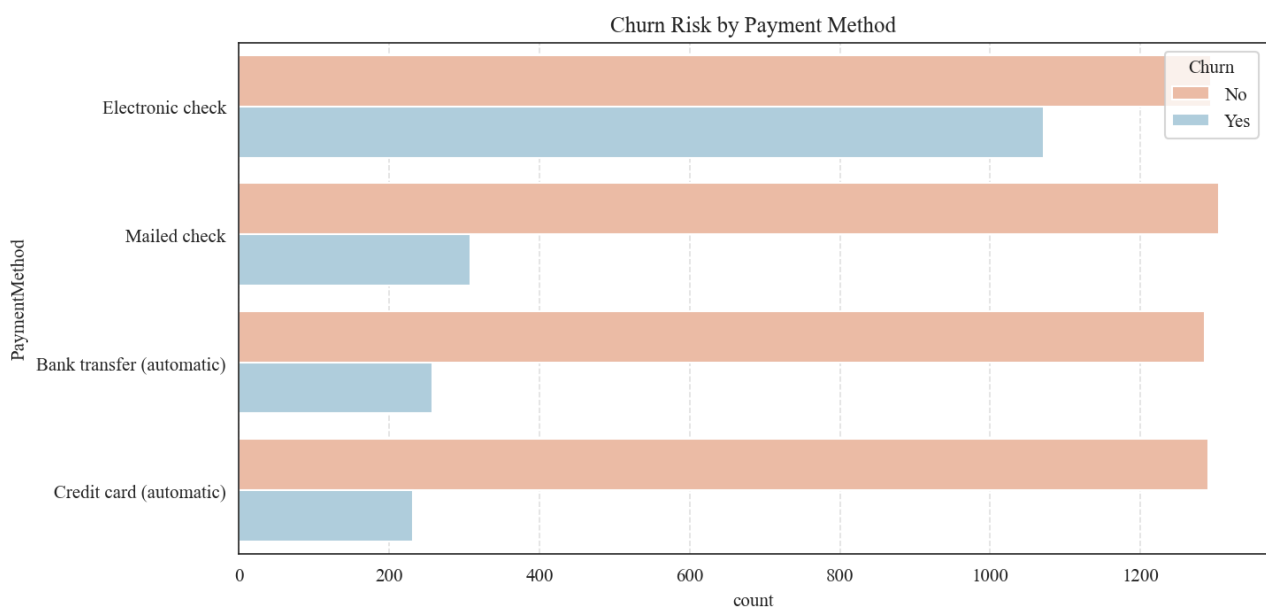
Analysis: This is the most critical finding. Month-to-Month contracts have the highest churn. Senior citizens also show a preference for these shorter, more flexible contracts, which aligns with their higher churn rate.

18. Paperless Billing



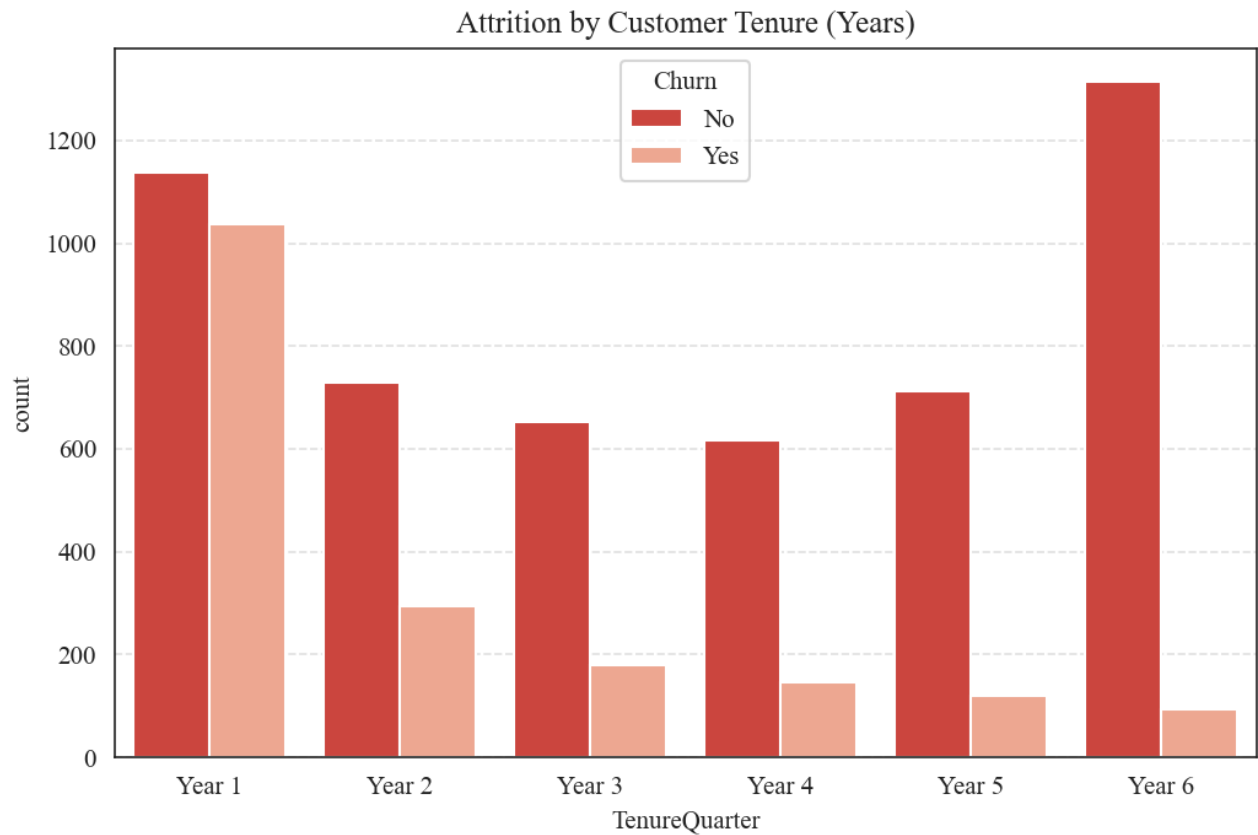
Analysis: Customers using Paperless Billing are more prone to churn. This correlates with a more digitally savvy user base that can easily switch providers.

19. Payment Methods



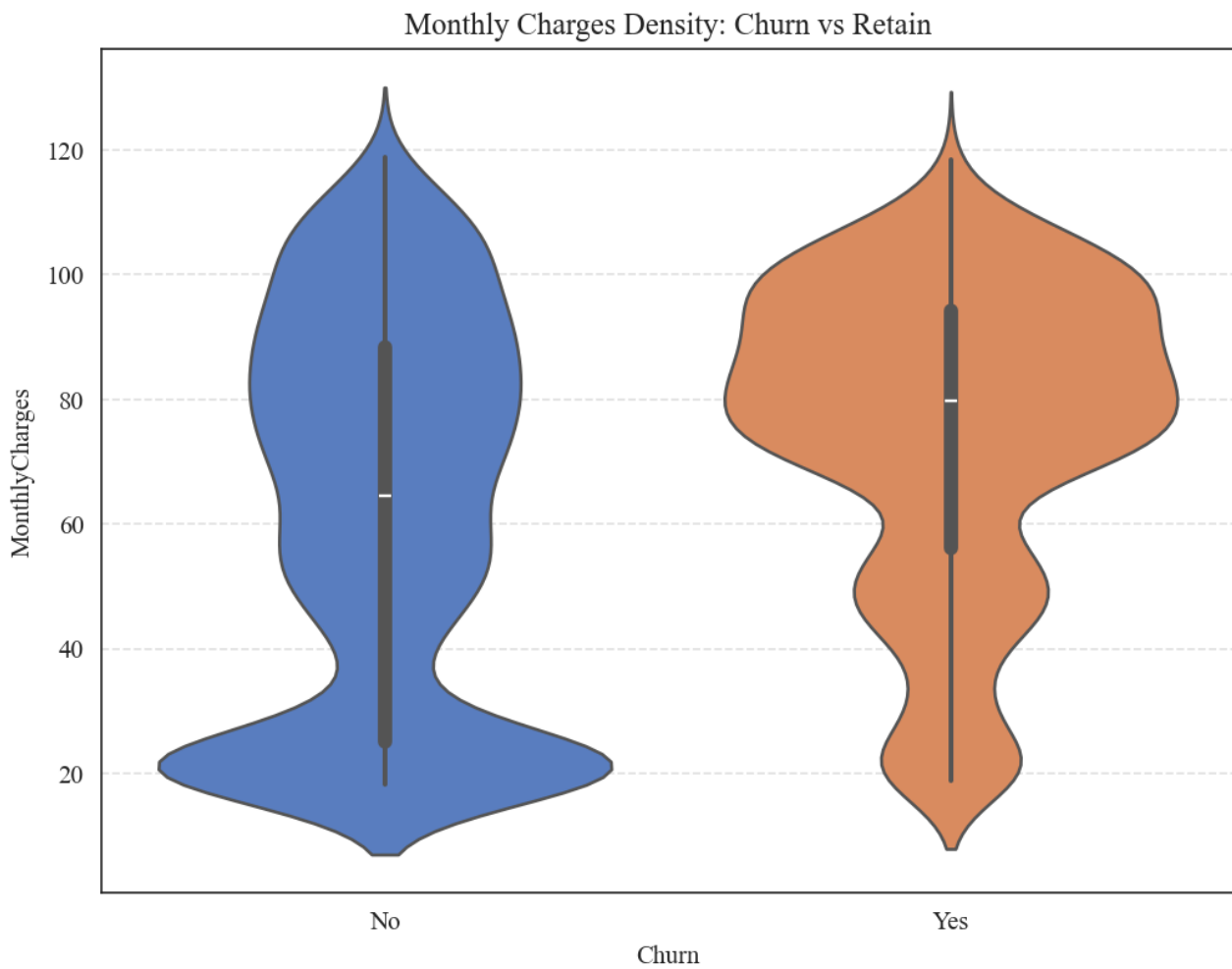
Analysis: Electronic Checks are associated with the highest churn. In contrast, automated Credit Card payments generally denote higher loyalty.

20. Tenure (Time with Company)



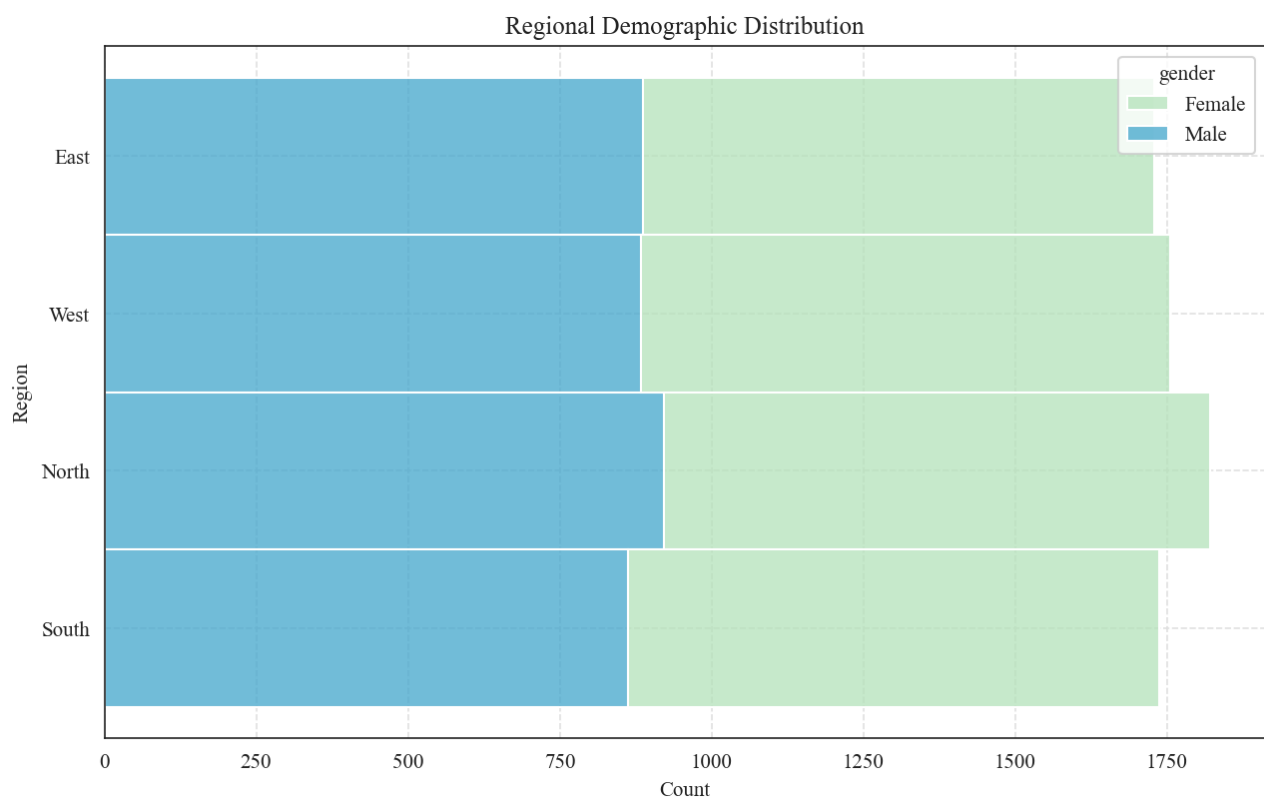
Analysis: The 'Year 1' cohort (Tenure < 12 months) faces the steepest attrition. Survival analysis suggests that if a customer stays beyond the first year, they are likely to remain for the long term.

21. Monthly Charges



Analysis: The violin plot shows a 'fat tail' at higher costs for churned customers. Price sensitivity is a major churn driver.

22. Region Analysis



Analysis: Regional distribution is relatively uniform, suggesting that churn is a systemic issue rather than a localized one.

5. Conclusion

Following this rigorous visual analysis, we recommend the following strategic initiatives:

- **Contract Migration:** Aggressively market 1-Year and 2-Year contracts with incentives to reduce the volatility of the Month-to-Month base.
- **Fiber Optic Review:** Conduct a technical and pricing audit of the Fiber Optic service to address the disproportionately high churn in this premium segment.
- **Onboarding Focus:** Implement a 'First Year' care program to support new customers through the high-risk retention period.
- **Senior Support:** Develop specific low-cost, flexible plans with dedicated support lines for Senior Citizens to reduce their attrition.